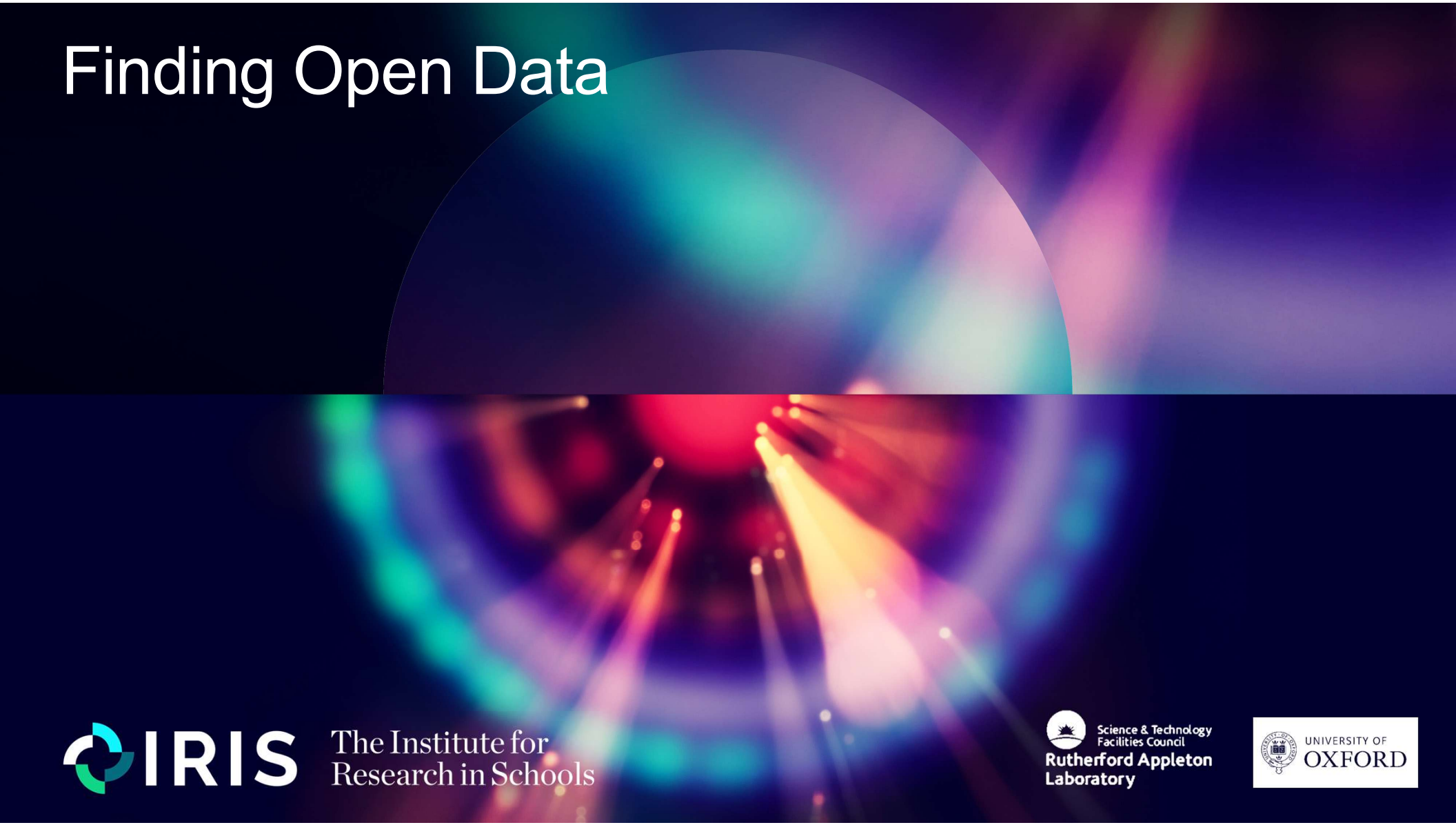
Finding Open Data



IRIS The Institute for Research in Schools

Science & Technology Facilities Council
Rutherford Appleton Laboratory

UNIVERSITY OF OXFORD

Finding Open Data

First, go to the ATLAS Open Data website,
https://opendata.atlas.cern/

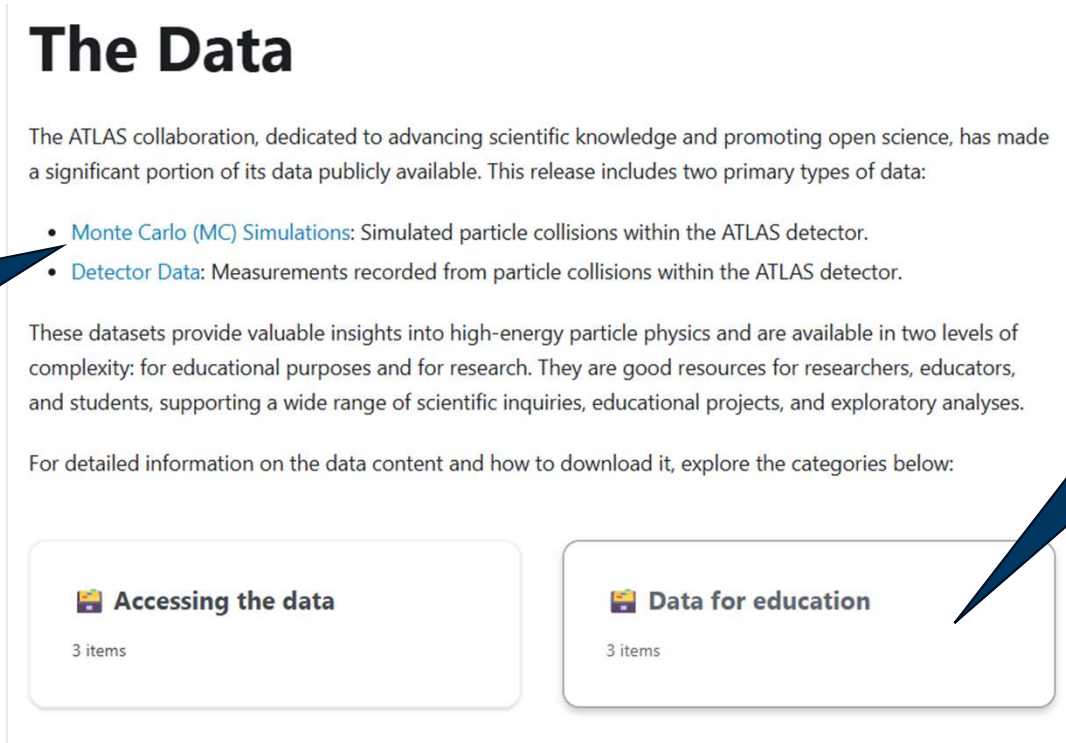
Finding Open Data



High Energy Physics data for everyone.

Finding Open Data

Read through the information linked here.



The Data

The ATLAS collaboration, dedicated to advancing scientific knowledge and promoting open science, has made a significant portion of its data publicly available. This release includes two primary types of data:

- Monte Carlo (MC) Simulations: Simulated particle collisions within the ATLAS detector.
- Detector Data: Measurements recorded from particle collisions within the ATLAS detector.

These datasets provide valuable insights into high-energy particle physics and are available in two levels of complexity: for educational purposes and for research. They are good resources for researchers, educators, and students, supporting a wide range of scientific inquiries, educational projects, and exploratory analyses.

For detailed information on the data content and how to download it, explore the categories below:

Accessing the data

3 items

Data for education

3 items

Then click on Data for Education

Finding Open Data

Data for education

We currently provide educational datasets from two proton-proton collision runs: one recorded at a center-of-mass energy of **8 TeV** using data collected in **2012**, and one recorded at **13 TeV** using data collected in **2016**. These datasets are designed to support teaching and learning across a range of analysis topics, from introductory studies to more advanced classroom exercises. The information for each collection is summarized in the table below.

Year	Release	Events [M]	Datasets	Volume [GB]
2016	8 TeV	59.990	44	6.384
2020	13 TeV	942.4	228	71.4
2025	13TeV beta	9838	374	2500

Click on the following cards to explore each of the releases.

13 TeV 2025 Data

1 item

13 TeV 2020 Data

2 items

8 TeV Data

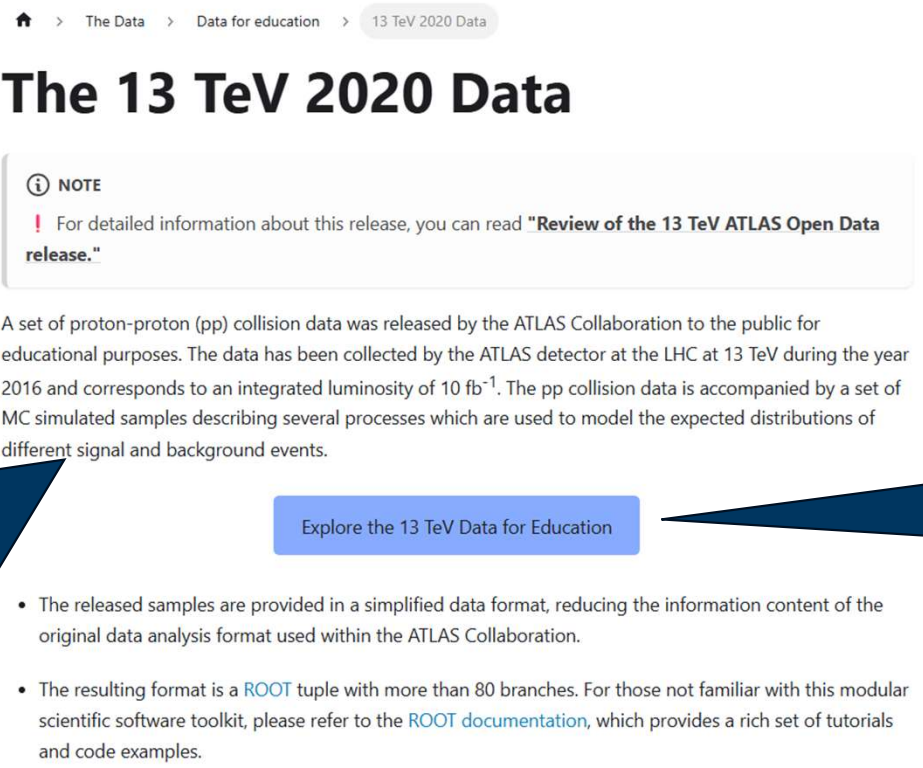
We will initially be using the 2020 data.

If you want to you can explore other data sets when conducting your own research in Phase 3

Finding Open Data

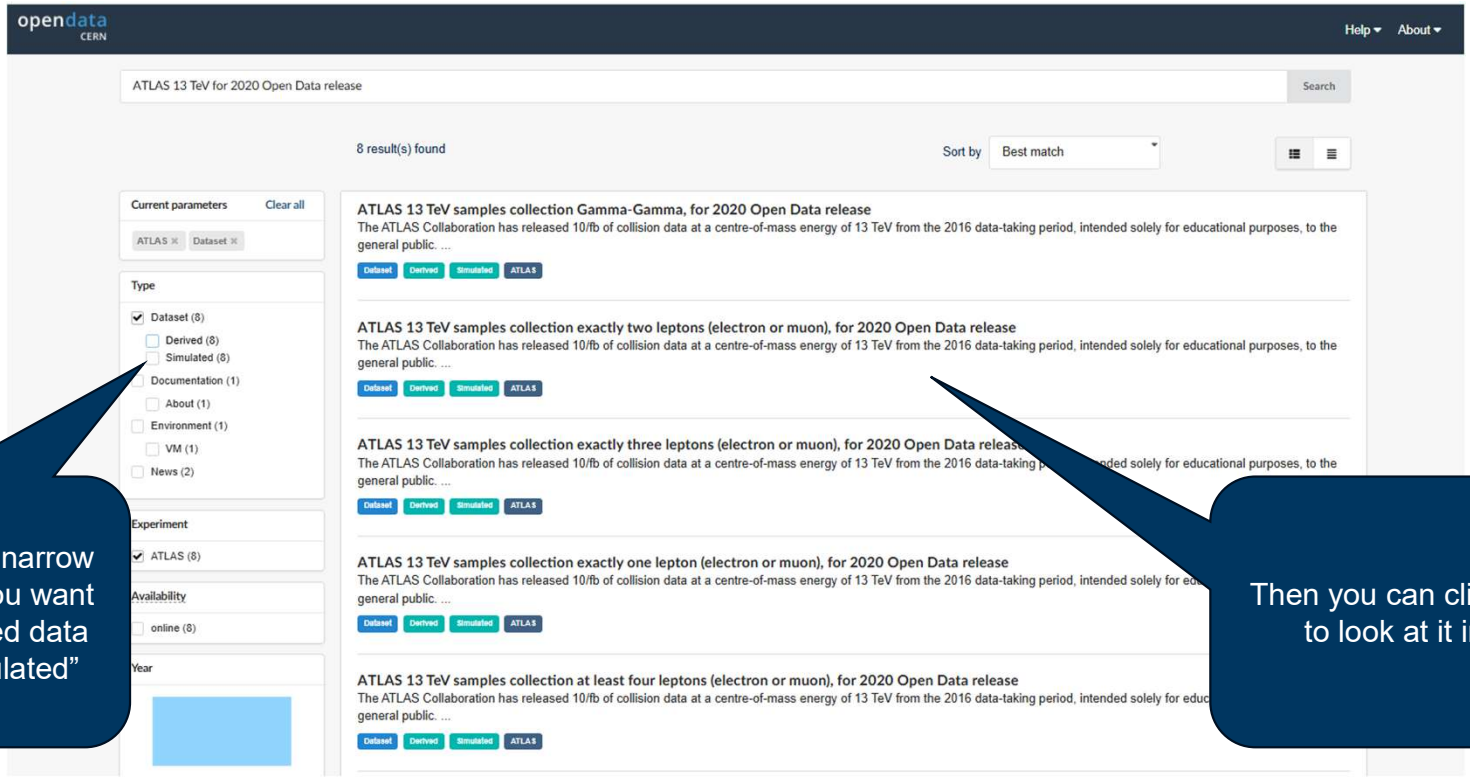
Read down through this page. Some parts might be confusing, but you will gradually develop familiarity with the notation and the contents of the files.

You will likely revisit it and the documentation tab a few times to read about the different types of analysis you can do and to find the numbers/names of datafiles for those analyses.



When you're ready, click the blue button to navigate to the files themselves.

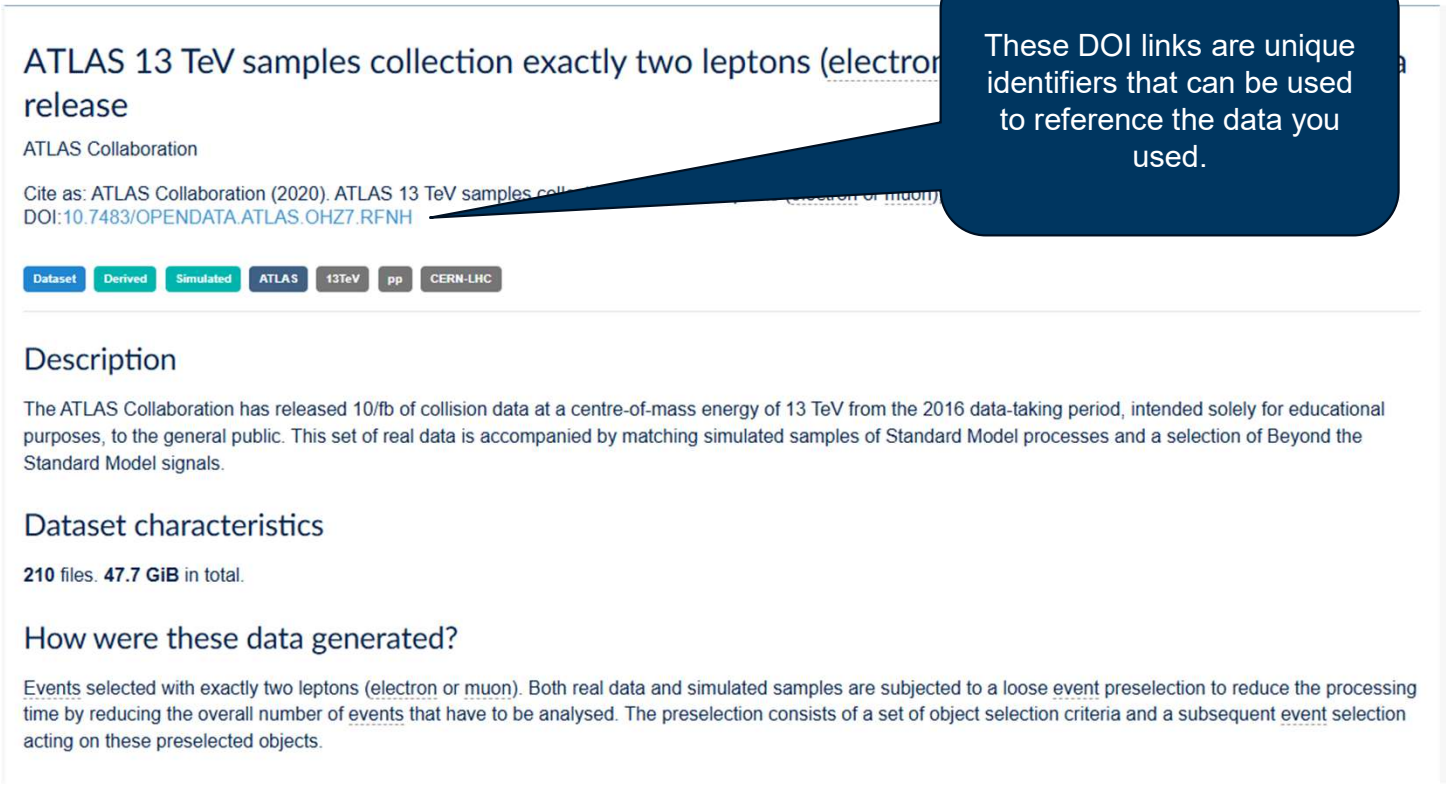
Finding Open Data



You can use filters to narrow your search. E.g. if you want Monte Carlo simulated data you can select “simulated”

Then you can click on a row to look at it in detail.

Finding Open Data



ATLAS 13 TeV samples collection exactly two leptons (electron
release

ATLAS Collaboration

Cite as: ATLAS Collaboration (2020). ATLAS 13 TeV samples
DOI:10.7483/OPENDATA.ATLAS.OHZ7.RFNH

Dataset | Derived | Simulated | ATLAS | 13TeV | pp | CERN-LHC

Description

The ATLAS Collaboration has released 10/fb of collision data at a centre-of-mass energy of 13 TeV from the 2016 data-taking period, intended solely for educational purposes, to the general public. This set of real data is accompanied by matching simulated samples of Standard Model processes and a selection of Beyond the Standard Model signals.

Dataset characteristics

210 files. **47.7 GiB** in total.

How were these data generated?

Events selected with exactly two leptons (electron or muon). Both real data and simulated samples are subjected to a loose event preselection to reduce the processing time by reducing the overall number of events that have to be analysed. The preselection consists of a set of object selection criteria and a subsequent event selection acting on these preselected objects.

Finding Open Data

This dataset is provided by the ATLAS Collaboration **only for educational purposes and is not suited for scientific publications**. You can access these data through the ATLAS Virtual Machine and Software. More documentation is available at:

The ATLAS Open Data documentation for 13 TeV release

The ATLAS Open Data site

Files and indexes

Filename	Size	
data_A.exactly2lep.root	122.1 MiB	Download
data_B.exactly2lep.root	450.8 MiB	Download
data_C.exactly2lep.root	657.0 MiB	Download
data_D.exactly2lep.root	1009.4 MiB	Download
exactly2lep.zip	20.9 GiB	Download

« ‹ 1 2 3 4 5 ... 42 › »

There are links to the documentation

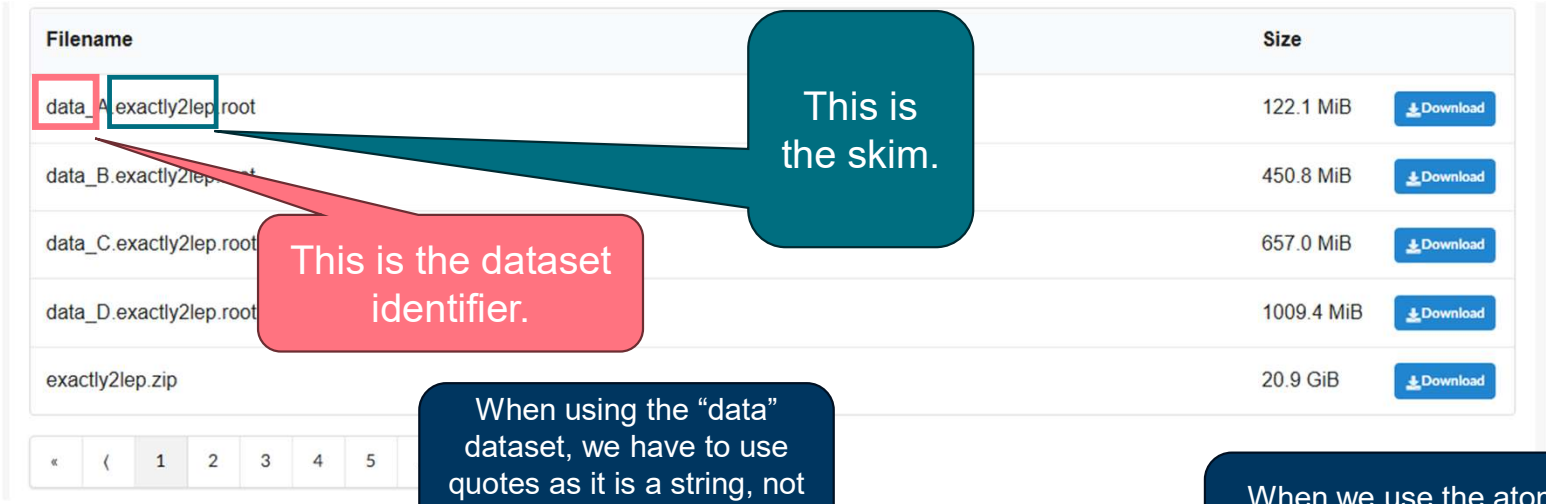
It is possible to download the files and work with them locally, but as they are very large this is inefficient. It also doesn't work well with the Monty server.

Sometimes there are a lot of files!

Finding Open Data



Finding Open Data



When using the "data" dataset, we have to use quotes as it is a string, not an integer (number)

When we use the atom.get_urls() attribute we can see the data_A, data_B, data_C and data_D results, as expected.

```
atom.get_urls("data", skim="exactly2lep")
```

```
['root://eospublic.cern.ch:1094//eos/opendata/atlas/OutreachDatasets/2020-08-19/exactly2lep/Data/data_A.exactly2lep.root',
 'root://eospublic.cern.ch:1094//eos/opendata/atlas/OutreachDatasets/2020-08-19/exactly2lep/Data/data_B.exactly2lep.root',
 'root://eospublic.cern.ch:1094//eos/opendata/atlas/OutreachDatasets/2020-08-19/exactly2lep/Data/data_C.exactly2lep.root',
 'root://eospublic.cern.ch:1094//eos/opendata/atlas/OutreachDatasets/2020-08-19/exactly2lep/Data/data_D.exactly2lep.root']
```

Finding Open Data

```
atom.get_urls(363491, skim="1largeRjet1lep")
```

```
['root://eospublic.cern.ch:1094//eos/opendata/atlas/OutreachDatasets/2020-08-19/1largeRjet1lep/MC/mc_363491.lllv.1largeRjet1lep.root']
```

So, to access the data we choose the first (and only!) result in the list ([0]) and use the syntax below (see more in the "AccessingRootFiles" notebook):

```
events_using_atom = uproot.open(atom.get_urls(363491, skim="1largeRjet1lep")[0])
```

```
atom.get_urls("data", skim="exactly2lep")
```

```
['root://eospublic.cern.ch:1094//eos/opendata/atlas/OutreachDatasets/2020-08-19/exactly2lep/Data/data_A.exactly2lep.root',
 'root://eospublic.cern.ch:1094//eos/opendata/atlas/OutreachDatasets/2020-08-19/exactly2lep/Data/data_B.exactly2lep.root',
 'root://eospublic.cern.ch:1094//eos/opendata/atlas/OutreachDatasets/2020-08-19/exactly2lep/Data/data_C.exactly2lep.root',
 'root://eospublic.cern.ch:1094//eos/opendata/atlas/OutreachDatasets/2020-08-19/exactly2lep/Data/data_D.exactly2lep.root']
```

So we can see it is worth having a look to check how many files there are!
To access data in a particular file, you can work your way down the list:
events_using_atom = uproot.open(atom.get_urls("data", skim="exactly2lep")[0]) – gives data_A
events_using_atom = uproot.open(atom.get_urls("data", skim="exactly2lep")[1]) – gives data_B
events_using_atom = uproot.open(atom.get_urls("data", skim="exactly2lep")[2]) – gives data_C
events_using_atom = uproot.open(atom.get_urls("data", skim="exactly2lep")[3]) - gives data_D

Then:
sel_atom_events = events_using_atom['mini'].arrays(["lep_n", "lep_charge", "lep_type", "lep_pt", "lep_eta", "lep_phi", "lep_E"])

Alternatively, if you know for sure that you want all the data there are a few different ways to combine the files. This is one way:
urls = atom.get_urls("data", skim="exactly2lep")
urls_with_tree = [f"{url}:mini" for url in urls]
all_events = uproot.concatenate(urls_with_tree, expressions = ["lep_n", "lep_charge", "lep_type", "lep_pt", "lep_eta", "lep_phi", "lep_E"])